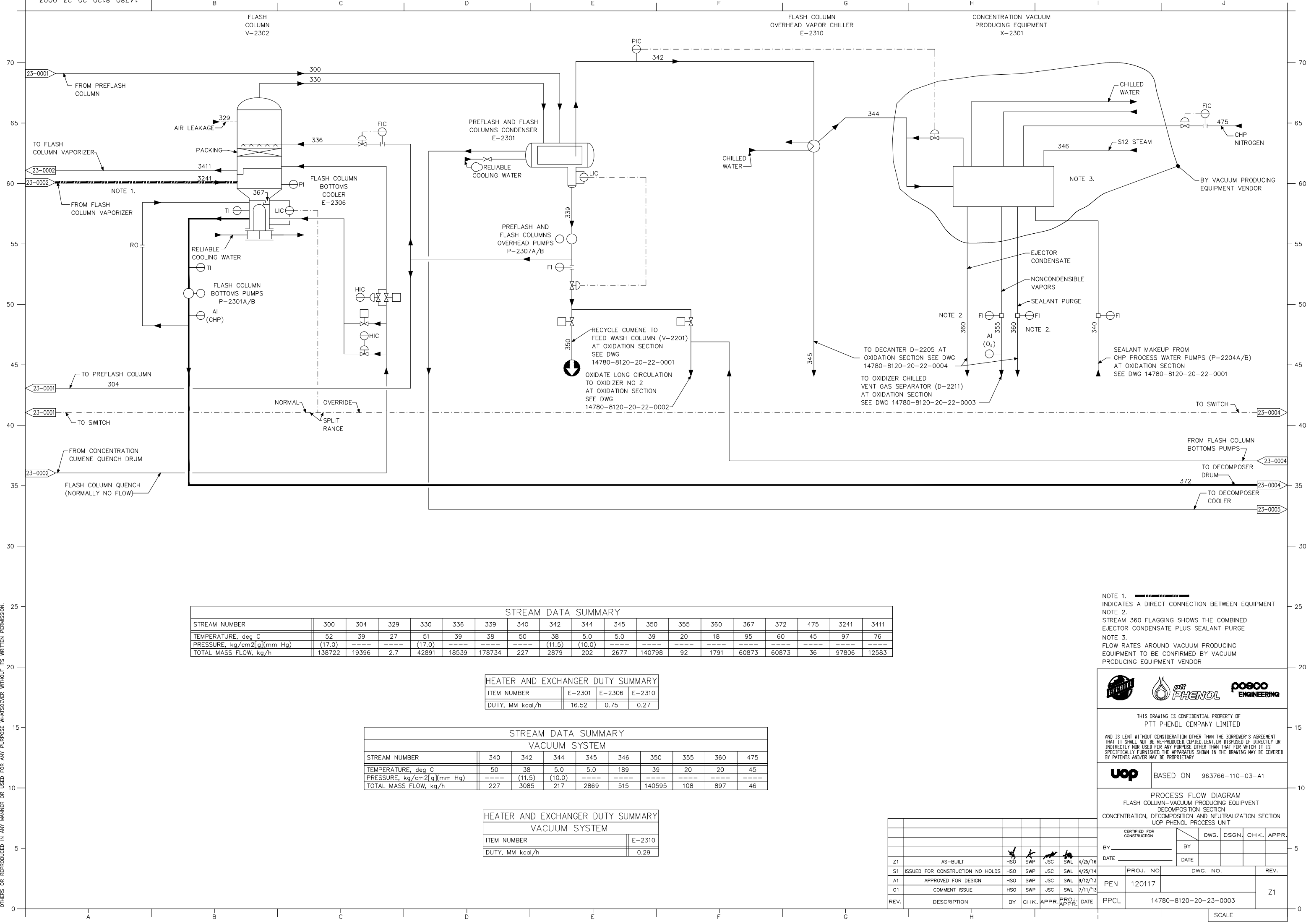


NOTE: THE INFORMATION IN THIS DOCUMENT IS CONFIDENTIAL AND THE PROPERTY OF UOP LLC AND MUST NOT BE DISCLOSED TO OTHERS OR REPRODUCED IN ANY MANNER OR USED FOR ANY PURPOSE WHATSOEVER WITHOUT ITS WRITTEN PERMISSION.



STREAM DATA SUMMARY																		
STREAM NUMBER	300	304	329	330	336	339	340	342	344	345	350	355	360	367	372	475	3241	3411
TEMPERATURE, deg C	52	39	27	51	39	38	50	38	5.0	5.0	39	20	18	95	60	45	97	76
PRESSURE, kg/cm2[g](mm Hg)	(17.0)	----	----	(17.0)	----	----	----	(11.5)	(10.0)	----	----	----	----	----	----	----	----	----
TOTAL MASS FLOW, kg/h	138722	19396	2.7	42891	18539	178734	227	2879	202	2677	140798	92	1791	60873	60873	36	97806	12583

HEATER AND EXCHANGER DUTY SUMMARY			
ITEM NUMBER	E-2301	E-2306	E-2310
DUTY, MM kcal/h	16.52	0.75	0.27

STREAM DATA SUMMARY									
VACUUM SYSTEM									
STREAM NUMBER	340	342	344	345	346	350	355	360	475
TEMPERATURE, deg C	50	38	5.0	5.0	189	39	20	20	45
PRESSURE, kg/cm2[g](mm Hg)	----	(11.5)	(10.0)	----	----	----	----	----	----
TOTAL MASS FLOW, kg/h	227	3085	217	2869	515	140595	108	897	46

HEATER AND EXCHANGER DUTY SUMMARY	
VACUUM SYSTEM	
ITEM NUMBER	E-2310
DUTY, MM kcal/h	0.29

REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	PROJ. NO.	DWG. NO.	REV.
Z1	AS-BUILT	HSO	SWP	JSC	SWL	4/25/16		
S1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	4/25/14		
A1	APPROVED FOR DESIGN	HSO	SWP	JSC	SWL	9/12/13		
O1	COMMENT ISSUE	HSO	SWP	JSC	SWL	7/11/13		
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	PPCL	14780-8120-20-23-0003	Z1

NOTE 1. **-----** INDICATES A DIRECT CONNECTION BETWEEN EQUIPMENT
NOTE 2. STREAM 360 FLAGGING SHOWS THE COMBINED EJECTOR CONDENSATE PLUS SEALANT PURGE
NOTE 3. FLOW RATES AROUND VACUUM PRODUCING EQUIPMENT TO BE CONFIRMED BY VACUUM PRODUCING EQUIPMENT VENDOR



THIS DRAWING IS CONFIDENTIAL PROPERTY OF PTT PHENOL COMPANY LIMITED
AND IS LENT WITHOUT CONSIDERATION OTHER THAN THE BORROWER'S AGREEMENT THAT IT SHALL NOT BE RE-PRODUCED, COPIED, LENT, OR DISPOSED OF DIRECTLY OR INDIRECTLY NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS SPECIFICALLY FURNISHED. THE APPARATUS SHOWN IN THE DRAWING MAY BE COVERED BY PATENTS AND/OR MAY BE PROPRIETARY

uop BASED ON 963766-110-03-A1

PROCESS FLOW DIAGRAM
FLASH COLUMN-VACUUM PRODUCING EQUIPMENT
DECOMPOSITION SECTION
CONCENTRATION, DECOMPOSITION AND NEUTRALIZATION SECTION
UOP PHENOL PROCESS UNIT

CERTIFIED FOR CONSTRUCTION
BY: _____ DWG. DSGN. CHK. APPR.
DATE: _____ DATE: _____

SCALE